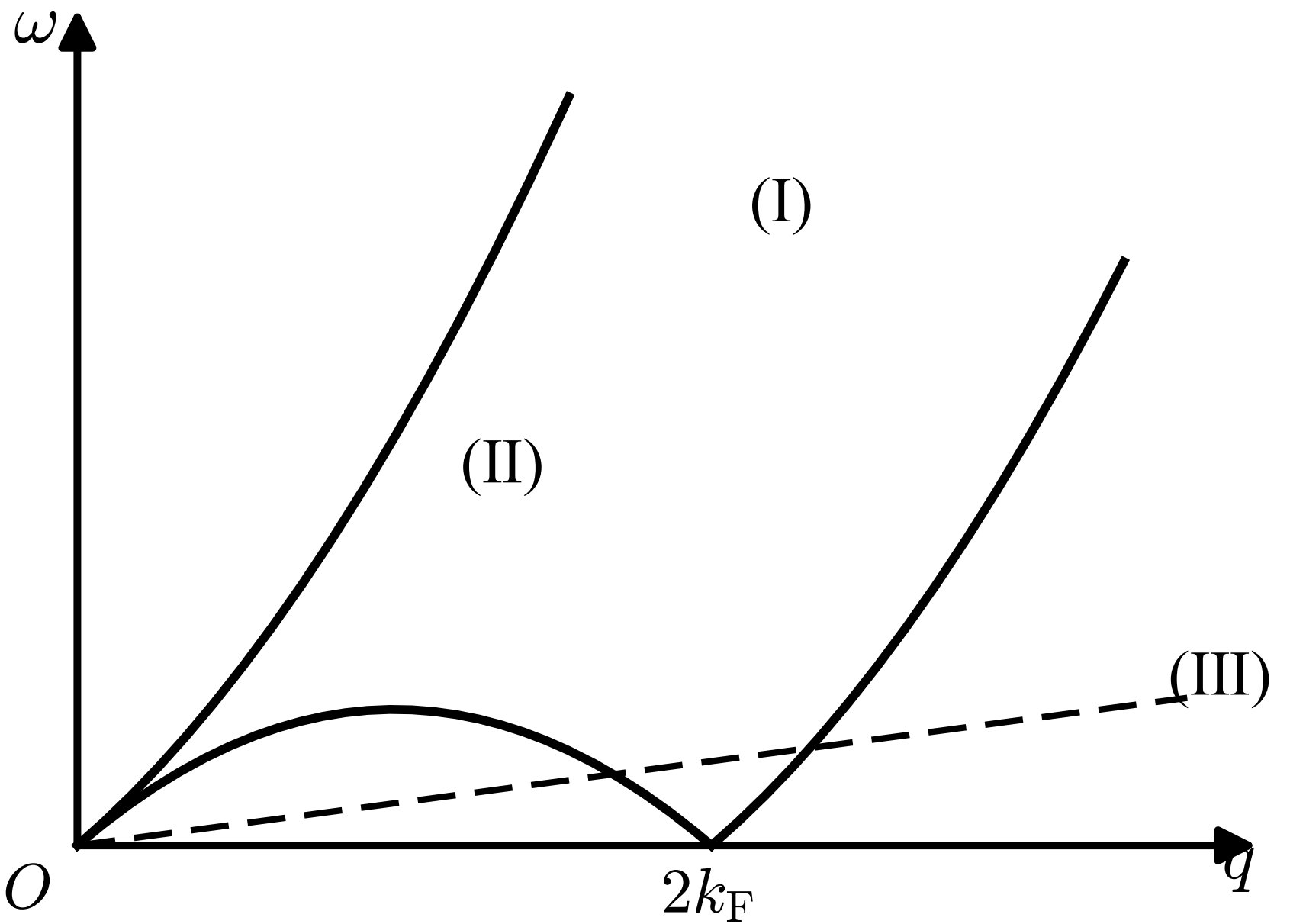
ω
(I)
(II)
(III)
O
$2k_{\mathrm{F}}$
q